TEXORA · LECTURE SHEET

Humidification

Textile fibres gain and lose moisture with the room air. Controlling humidity ratio and relative humidity is therefore a production control, not a comfort extra. This sheet covers definitions, system types and physical methods, dehumidification, cooling towers, and the psychrometric calculation every mill engineer must be able to do.

TE-4105 · INDUSTRIAL UTILITIES · TOPIC 1 · LEVEL 4 TERM 1

1. Definitions and basic quantities

Humidification is the controlled addition of water vapour to air so that the air reaches a specified moisture content. **Dehumidification** is the reverse — removal of water vapour. The state of moist air is fixed by dry-bulb temperature, wet-bulb temperature, relative humidity and humidity ratio.

- **Dry-bulb temperature (DBT)** — temperature measured by an ordinary thermometer.
- **Wet-bulb temperature (WBT)** — temperature measured with a wetted wick; evaporation cools it. DBT − WBT is the wet-bulb depression.
- **Relative humidity (RH)** — ratio of actual vapour pressure to saturation vapour pressure at the same temperature, expressed as a percentage: RH = p_v / p_s × 100.
- **Humidity ratio (W)** — mass of water vapour per unit mass of dry air, kg/kg or g/kg dry air: W = 0.622 p_v / (P − p_v), where P is total (atmospheric) pressure.
- **Dew point** — temperature at which the air becomes saturated on cooling at constant pressure and condensation begins.
- **Specific volume and enthalpy** — volume and total heat content per kg dry air, read from the psychrometric chart for fan and coil sizing.

Natural fibres are **hygroscopic**. Cotton regain rises with RH. Low RH causes brittleness, fly, static and ends down; excessive RH causes roller lapping, corrosion and mildew. Typical department standards (illustrative, mill practice varies) are: blowroom 55–60% RH, carding 50–55%, spinning 55–65%, weaving 70–80%.

2. Types of humidification system

TYPE	HOW IT WORKS	TYPICAL USE

Central air-washer plant	Air passes through a spray chamber; water evaporates into it; washed, humidified air is ducted to departments	Whole mill or large sheds; also cleans air
Unit / local humidifiers	Atomising units mounted in the department add mist directly to room air	Single department, supplementary control
Duct humidifiers	Spray or steam injected inside the supply duct	Trimming RH of centrally supplied air
Steam humidification	Live steam mixed into air stream	Small, precise or winter duties; energy-costly

A complete central plant normally contains: intake louvres and filters, spray chamber with nozzles and eliminator plates, recirculation pump and tank, fan, heating coil (winter), distribution ducts, and automatic RH sensors in each department controlling spray pumps and dampers.

3. Physical methods of humidification

3.1 Vaporisation

Water is boiled or heated so it enters the air as true vapour. Steam injection is the direct form. The air gains both moisture and sensible heat, so on the psychrometric chart the state moves up and slightly to the right. Control is precise and hygienic (no liquid droplets), but energy cost is the highest of the three methods.

3.2 Atomisation

Water is broken into a very fine mist by compressed-air nozzles, high-pressure nozzles or spinning discs, and the droplets evaporate in the room or duct. Because evaporation takes latent heat from the air, atomisation is approximately **adiabatic**: DBT falls slightly as W rises, and the process follows a near-constant wet-bulb line. Advantages are low energy use and local zoning; risks are nozzle blockage, mineral dust if water is untreated, and wetting if droplets are too coarse.

3.3 Evaporation

Air is passed over or through a wetted surface — spray chamber, wetted pads, or open water surface — and picks up moisture by evaporation. The air washer works this way: only part of the sprayed water evaporates (evaporation efficiency typically 60–80% for a single bank, illustrative), and the remainder is filtered, treated and recirculated. Evaporation also washes dust and fibre fly from the air, a major benefit in spinning mills.

> **KEY DISTINCTION**

> Vaporisation adds heat with the moisture. Atomisation and evaporation are essentially adiabatic — moisture rises at nearly constant wet-bulb temperature, so dry-bulb temperature falls a little unless a heating coil compensates. In the worked example below the small temperature change is neglected and the process is drawn at constant 30 °C for clarity of the humidity-ratio calculation; a full plant design would follow the wet-bulb line and add reheat if exact DBT must be held.

4. Psychrometric worked example — water to be added

Problem. Condition air at 30 °C from 40% RH to 65% RH. Find the humidity ratio before and after, the water to add per kg dry air, and the water rate for a dry-air flow of 15,000 kg/h. Atmospheric pressure P = 101.325 kPa.

Formulae used (stated in full). Saturation vapour pressure by the Magnus formula:

$$p_s = 0.61094 \times \exp[\, 17.625\, T / (T + 243.04)\,] \quad \text{kPa} \quad (T \text{ in } °C)$$

$$p_v = RH \times p_s \quad \cdot \quad W = 0.622\, p_v / (P - p_v)$$

Step 1 — saturation pressure at 30 °C. $p_s = 0.61094 \times \exp[17.625 \times 30 / (30 + 243.04)] = 0.61094 \times \exp(1.936) =$ **4.24 kPa**. This agrees with standard steam-table values at 30 °C (about 4.25 kPa).

Step 2 — before (40% RH). $p_{v1} = 0.40 \times 4.24 = 1.695$ kPa. $W_1 = 0.622 \times 1.695 / (101.325 - 1.695) = 1.054 / 99.63 = 0.01058$ kg/kg = **10.6 g/kg dry air**.

Step 3 — after (65% RH). $p_{v2} = 0.65 \times 4.24 = 2.754$ kPa. $W_2 = 0.622 \times 2.754 / (101.325 - 2.754) = 1.713 / 98.57 = 0.01738$ kg/kg = **17.4 g/kg dry air**.

Step 4 — water to add. $\Delta W = 17.4 - 10.6 =$ **6.8 g per kg dry air**. For 15,000 kg dry air/h: water = $15{,}000 \times 0.0068 =$ **102 kg/h** must actually evaporate. If the spray evaporation efficiency is 70%, the plant must spray $102 / 0.70 \approx$ **146 kg/h**, the rest draining for recirculation.

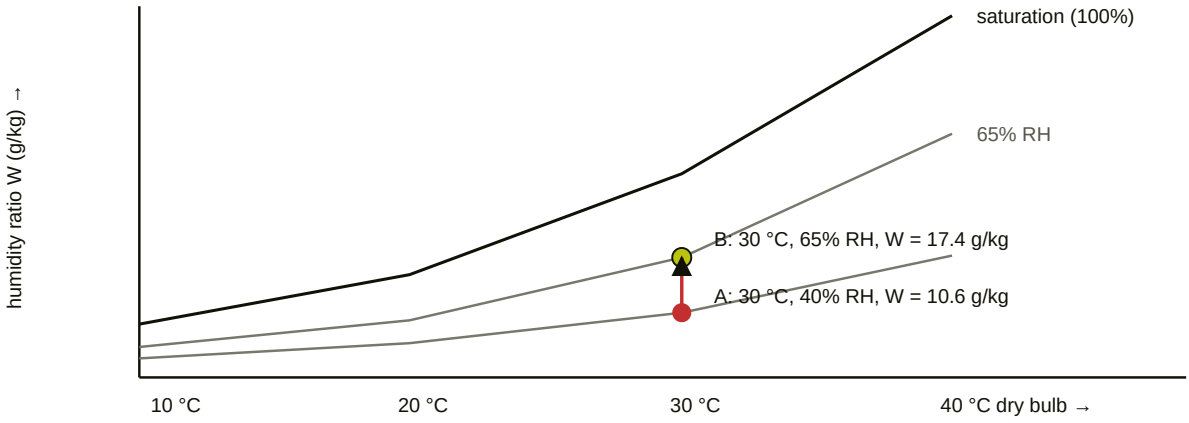


Figure 1 — Psychrometric chart (sketch). Humidification A → B at 30 °C: W rises from 10.6 to 17.4 g/kg, ΔW = 6.8 g/kg, exactly the values of the worked example. Curves are schematic; values are calculated, not read by eye.

5. Dehumidifier

A **dehumidifier** removes moisture when RH is too high — in monsoon weather, in warehouses, and around moisture-sensitive stores. Two principles dominate:

- **Refrigeration (condensing) type:** air is cooled below its dew point over a refrigerated coil, water condenses and drains, and the air is reheated slightly by the condenser. Effective in warm, humid air.
- **Desiccant type:** air passes through a moisture-absorbing material (silica gel, lithium chloride, desiccant wheel). The desiccant is regenerated with heated air. Effective at low temperatures and for very low dew points.

In a textile mill dehumidification protects stored yarn and cloth from mildew, keeps sizing ingredients dry, and prevents condensation and corrosion in idle departments.

6. Cooling tower

Many mill services — air-washer recirculation, compressors, chillers, dye-house heat exchangers — reject heat to water. A **cooling tower** cools that warm water by evaporating a small part of it into a stream of air, so the water can be reused instead of wasted.

Working principle, step by step: (1) Warm water from the plant enters the top of the tower. (2) Spray nozzles distribute it over **fill (packing)** — a large wetted surface of slats or honeycomb. (3) Air is drawn upward (induced-draught fan) or forced upward (forced-draught fan), counter-current to the falling water. (4) About 1–2% of the water (typical, illustrative) evaporates; the latent heat comes from the remaining water, which cools by typically 5–10 °C. (5) Cooled water collects in the basin and

returns to the plant. Drift eliminators reduce droplet carry-over; make-up water replaces evaporation, drift and blowdown.

Key terms: *range* = hot-water temperature minus cold-water temperature; *approach* = cold-water temperature minus ambient wet-bulb temperature. A tower can never cool water below the ambient wet-bulb temperature. **Types:** natural-draught (hyperbolic, large power stations) and mechanical-draught (induced or forced, used in mills); cross-flow and counter-flow arrangements.

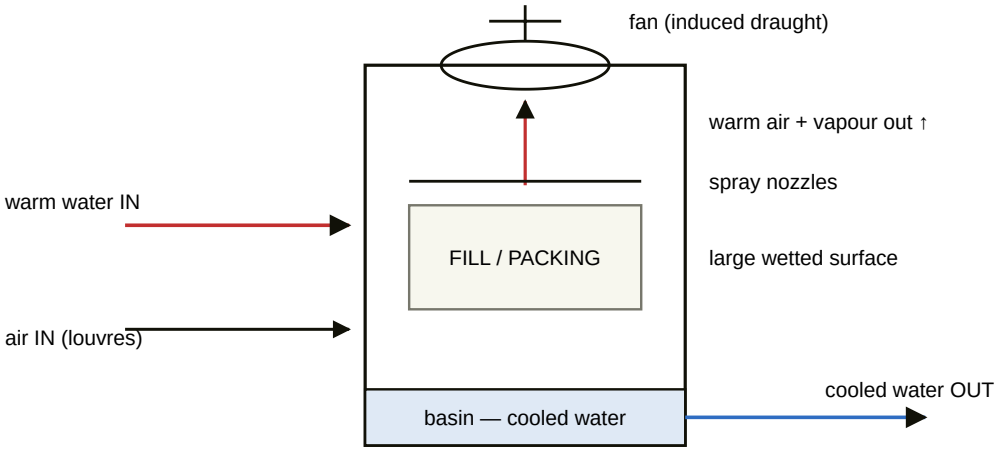


Figure 2 — Induced-draught counter-flow cooling tower. Warm water is sprayed over the fill, air rises against it, a small fraction evaporates, and cooled water collects in the basin for reuse.

7. Application to the textile industry

- **Spinning:** stable RH keeps regain, strength and flexibility constant; ends down, fly and static fall; count variation from moisture change is reduced.
- **Weaving:** the highest RH in the mill (about 70–80%, typical) keeps sized warp flexible and reduces warp breaks.
- **Weaving and knitting yarn stores:** conditioning rooms hold yarn at standard regain before dispatch so invoice weight is correct.
- **Wet processing:** controlled humidity in drying and finishing ranges gives uniform moisture in delivered cloth.
- **Testing laboratories:** standard atmosphere (commonly 20 ± 2 °C, 65 ± 2% RH in textile testing standards) makes test results comparable worldwide.
- **Worker comfort and dust control:** washed, humidified air carries less floating fly and dust.

Glossary

Humidification — controlled addition of water vapour to air. **Humidity ratio (W)** — kg water vapour per kg dry air. **Relative humidity** — actual vapour pressure as % of saturation pressure at the same temperature. **Wet-bulb temperature** — temperature shown by a wetted thermometer in moving air. **Atomisation** — breaking water into fine droplets that evaporate in air. **Air washer** — spray chamber that humidifies and cleans air. **Desiccant** — a material that absorbs moisture and can be regenerated by heating. **Cooling tower** — structure that cools recirculating water by partial evaporation. **Approach** — cooled-water temperature minus ambient wet-bulb temperature.

Key points

• $W = 0.622\ p_v/(P - p_v)$; at 30 °C, $p_s = 4.24$ kPa by the Magnus formula.

• Worked example: 30 °C, 40% → 65% RH raises W from 10.6 to 17.4 g/kg; ΔW = 6.8 g/kg; at 15,000 kg dry air/h, 102 kg/h must evaporate.

• Vaporisation adds heat; atomisation and evaporation are near-adiabatic.

• Dehumidifiers work by condensation below the dew point or by desiccant absorption.

• A cooling tower cools water by evaporating a small fraction of it; it cannot cool below ambient wet-bulb temperature.

Practice

1. Using the Magnus formula, calculate p_s at 25 °C, then W at 25 °C and 60% RH (P = 101.325 kPa).
2. Air at 30 °C is humidified from 40% to 65% RH (W: 10.6 → 17.4 g/kg). For a dry-air flow of 8,000 kg/h, find the evaporation rate required and the spray rate at 65% evaporation efficiency.
3. Explain why a cooling tower cannot deliver water colder than the wet-bulb temperature of the outside air, and define range and approach.
4. State two mill problems caused by RH that is (a) too low and (b) too high.

Answer key

1. $p_s = 0.61094 \times \exp[17.625 \times 25/(25 + 243.04)] = 0.61094 \times \exp(1.644) \approx$ **3.17 kPa**. $p_v = 0.60 \times 3.17 = 1.90$ kPa. $W = 0.622 \times 1.90/(101.325 - 1.90) \approx$ **0.0119 kg/kg = 11.9 g/kg**.
2. ΔW = 6.8 g/kg. Evaporation = 8,000 × 0.0068 = **54.4 kg/h**. Spray = 54.4/0.65 ≈ **83.7 kg/h**.
3. Cooling is by evaporation into the air; evaporation stops when air at the water surface is saturated, which occurs at the wet-bulb temperature — the thermodynamic limit. Range = hot-water in minus cold-water out; approach = cold-water out minus ambient wet-bulb.
4. (a) Too low: fibre brittleness and ends down; static, fly and dust. (b) Too high: roller lapping and sticky fibre; corrosion, mildew and overweight packages.